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Ex1: $X_1, \dots, X_n \stackrel{iid}{\sim} \frac{1}{\pi\{1+(x-\theta)^2\}}, -x, \theta \in \mathbb{R}$

$S = (X_{(1)}, \dots, X_{(n)}), X_{(1)} < X_{(2)} < \dots < X_{(n)}$ is suff.

~~$X_{(n)}$ is suff.~~ Is it complete?

• ~~$g(X_{(i)} - X_{(j)})$~~ , $i \neq j$ is a bounded function.

Distribution

is free of θ

~~For completeness~~, $g(X_{(i)} - X_{(j)})$ will have a finite expectation and it will be a constant free from θ .

Let the constant be c . Then $E_\theta[g(X_{(i)} - X_{(j)}) - c] = 0 \forall \theta$.

But $g(X_{(i)} - X_{(j)}) - c$ isn't necessarily the 0 funcⁿ.

$\therefore S$ is NOT a complete statistic. It's an ancillary statistic.

- Let $X \sim P_\theta$, $V(X)$ is a stat. whose distⁿ doesn't depend on θ . $V(X)$ is called an ancillary statistic.
- ~~Functions~~ A function g is called the zero function if $E_\theta(g(X)) = 0 \forall \theta \in \Theta$.

* Exponential Families of Distributions

Let $X \sim f_\theta(x)$ or $f(x|\theta) \leftarrow$ pdf or pmf !
They mean the same

f is said to be a member of an exponential family if:

(i) the support A of f doesn't depend on θ .

(ii) $f(x|\theta) = b(\theta) \exp\{c(\theta) \cdot T(x)\} h(x)$, where
 $b(\theta) > 0$, $h(x) > 0$

Ex2: Let $X \sim \text{Bin}(n, \theta)$, $\theta \in (0, 1)$

$$b(\theta) = (1-\theta)^n, \quad c(\theta) = \ln\left(\frac{\theta}{1-\theta}\right), \quad T(x) = x, \quad h(x) = \binom{n}{x}$$

Ex3: Let $X \sim \text{Poi}(\theta)$, $\theta > 0$

$$b(\theta) = e^{-\theta}, \quad h(x) = \frac{1}{x!}, \quad c(\theta) = \ln \theta, \quad T(x) = x$$

Ex4: Let $X \sim N(\theta, 1)$, $\theta \in \mathbb{R}$

$$b(\theta) = \frac{e^{-\theta^2/2}}{\sqrt{2\pi}}, \quad h(x) = e^{-x^2/2}, \quad c(\theta) = \theta, \quad T(x) = x$$

Ex5: Let $X \sim \text{Cauchy}(\theta)$, $\theta \in \mathbb{R}$

Check why it's NOT in the exponential family

Ex6: Let $X \sim \frac{1}{\Gamma(\gamma)} \cdot \frac{1}{\theta^\gamma} \cdot \exp\left(\frac{-x}{\theta}\right) x^{\gamma-1}$, $x, \theta > 0$
 $\downarrow$
 known ($\gamma > 0$)

$$p(\theta) = (\Gamma(\gamma) \cdot \theta^\gamma)^{-1}, \quad c(\theta) = -\frac{1}{\theta}, \quad T(x) = x, \quad h(x) = x^{\gamma-1}$$

→ An interesting yet simple result:

Let $X_1, \dots, X_n \stackrel{iid}{\sim} f(x|\theta)$, and f belongs to some exponential family.

Joint pdf/pmf:

$$f(\underline{x}|\theta) = (p(\theta))^n \cdot \prod_{i=1}^n h(x_i) \exp \left\{ c(\theta) \cdot \sum_{i=1}^n T(x_i) \right\}$$

Using NF factorization, $\sum_{i=1}^n T(x_i)$ is suff. Let $S = \sum_{i=1}^n T(x_i)$.

- Is it complete?

~~Let g be some funcⁿ s.t. $E(g(S)) = 0$ ($S = \sum_{i=1}^n T(x_i)$)~~

- Assume $c(\theta)$ to be a 1-1 funcⁿ of $\theta \in \mathcal{H}$.

Call $\eta = c(\theta)$ the natural parameter.

$\mathcal{H}$ is called the original parameter space.

$\eta \in \Gamma \rightarrow$ it's called the natural parameter space.

Result: If the natural parameter space Γ contains an interval w/ a non-empty interior the suff. stat.

$S = \sum_{i=1}^n T(x_i)$ is complete.

* A family of distributions: (Power Series Distribution)

$$f(x|\theta) = \text{pmf} = \frac{a(x) \cdot \theta^x}{g(\theta)}, \quad \theta \in (0, b) \quad (b \text{ can be } \infty)$$

$$g(\theta) > 0, a(x) \geq 0$$

$$x = 0, 1, \dots, n, \dots \quad \& \quad \sum_{x=0}^{\infty} a(x) \cdot \theta^x = g(\theta) > 0$$

→ ~~Ex 1~~ $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x|\theta) \leftarrow 0 < \theta < b$ a power series distribution

By NF-factorizⁿ, $S = \sum_{i=1}^n X_i$ is suff.

$$f(x|\theta) = a(x) \cdot \theta^x \cdot (g(\theta))^{-1}$$

$$= (g(\theta))^{-1} \cdot \exp(x \cdot \ln \theta) \cdot a(x)$$

$\therefore f$ belongs in exponential family.

We see, here $\Gamma = (-\infty, \ln b)$ & it contains an interval. $\therefore S$ is complete.

→ An example: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$ $\theta \in \mathbb{R}, \sigma^2 > 0$

~~unknown~~ → known

$\bar{X} = \sum_{i=1}^n X_i / n$ is complete sufficient for θ .

$S^2 = \sum_{i=1}^n (X_i - \bar{X})^2$ is an ancillary statistic. (Its distⁿ is free of θ).

- We know, in this case $\bar{X}$ & S^2 are indep. Also $\bar{X}$ is complete suff. & S^2 is ancillary. Could there be a connection b/w these 2 things.

Hint: $T = \bar{X}$ is a complete suff. stat.

$S = S^2$ is an ancillary stat.

$$P[S \leq s | T] = g(T)$$

$$E_\theta[g(T)] = P[S \leq s] \quad \forall \theta$$

$$\Rightarrow E_\theta[g(T) - P[S \leq s]] = 0 \quad \forall \theta$$

$$\Rightarrow g(T) = P[S \leq s] \quad \forall s \quad (\because T \text{ is complete})$$

$$\therefore P[S \leq s | T] = P[S \leq s] \quad \forall s$$

$\Rightarrow S$ & T are independent.

This result is known as Basu's theorem:

- "An ancillary statistic is always independent of a complete ~~suff~~ sufficient statistic in a given problem".

* Let $X_1, X_2, \dots$ $\stackrel{iid}{\sim} P_\theta$

For every $n \geq 1$ let $T_n = T_n(X_1, \dots, X_n)$ be an estimate for $\psi(\theta)$. We say that T_n is consistent (weakly consistent) for $\psi(\theta)$ if $T_n \xrightarrow[n \rightarrow \infty]{P} \psi(\theta)$

→ ~~Let~~ Let $T_n(X_1, \dots, X_n)$ be the respective UMVUE's for $\psi(\theta)$.

Then $T_n \xrightarrow[n \rightarrow \infty]{P} \psi(\theta)$.

Prove this.